

TASK : #51631

GOAL: Enable file transaction to servers using file transfer protocol[FTP]

1. configure sftp in an instance such that, it can receive files via port number 22 to the specific directory as the specific user. ie, the sftp users have restricted privileges.
2. verify the working with any FTP client like filezilla, cyberduck etc.
3. install and configure vsftpd in an instance such that, it can receive files very securely via port 21 to the configured directory as configured user. for Vsftp also privileges are restricted.
4. verify the working with any FTP client.

Step 1: Create the SFTP user with no shell access

```
#sudo useradd /home/sftpuser  
#sudo usermod -s /sbin/nologin sftpuser  
#sudo passwd sftpuser
```

-s /sbin/nologin — prevents shell login, only file transfer allowed

Step 2: Set up the chroot directory structure

SSH requires the **chroot root** to be owned by root and not writable by others.

```
#sudo mkdir -p /sftp/sftpuser  
#sudo chown root:root /sftp/sftpuser  
#sudo chmod 755 /sftp/sftpuser
```

Create a writable upload folder inside the chroot

```
#sudo mkdir -p /sftp/sftpuser/file  
#sudo chown sftpuser:sftpusers /sftp/sftpuser/uploads  
#sudo chmod 750 /sftp/sftpuser/uploads
```

The user will only be able to write inside uploads/ — the root of the chroot must remain root-owned or SSH will refuse to start the session.

Step 3: Configure sshd_config

```
#sudo vi /etc/ssh/sshd_config
```

Comment out or ensure this line exists (usually already present):

Subsystem sftp internal-sftp

Add this block at the **end of the file**:

```
Subsystem sftp internal-sftp  
Match User sftpuser  
  
    ForceCommand internal-sftp  
  
    PasswordAuthentication yes  
  
    ChrootDirectory /home/sftpuser
```

Match Group — applies these rules only to sftpusers group

ChrootDirectory /sftp/%u — jails user to their own folder (%u = username)

ForceCommand internal-sftp — forces SFTP subsystem only, blocks shell/SCP

AllowTcpForwarding no / X11Forwarding no — hardens against tunnel abuse

Step4: Add file to the directory

For that change the user to sftpuser:

```
#Sudo su – sftpuser
```

Then enter the password and login

Create a file :

```
#touch test  
  
#vi test
```

press I and write something

save and exit;

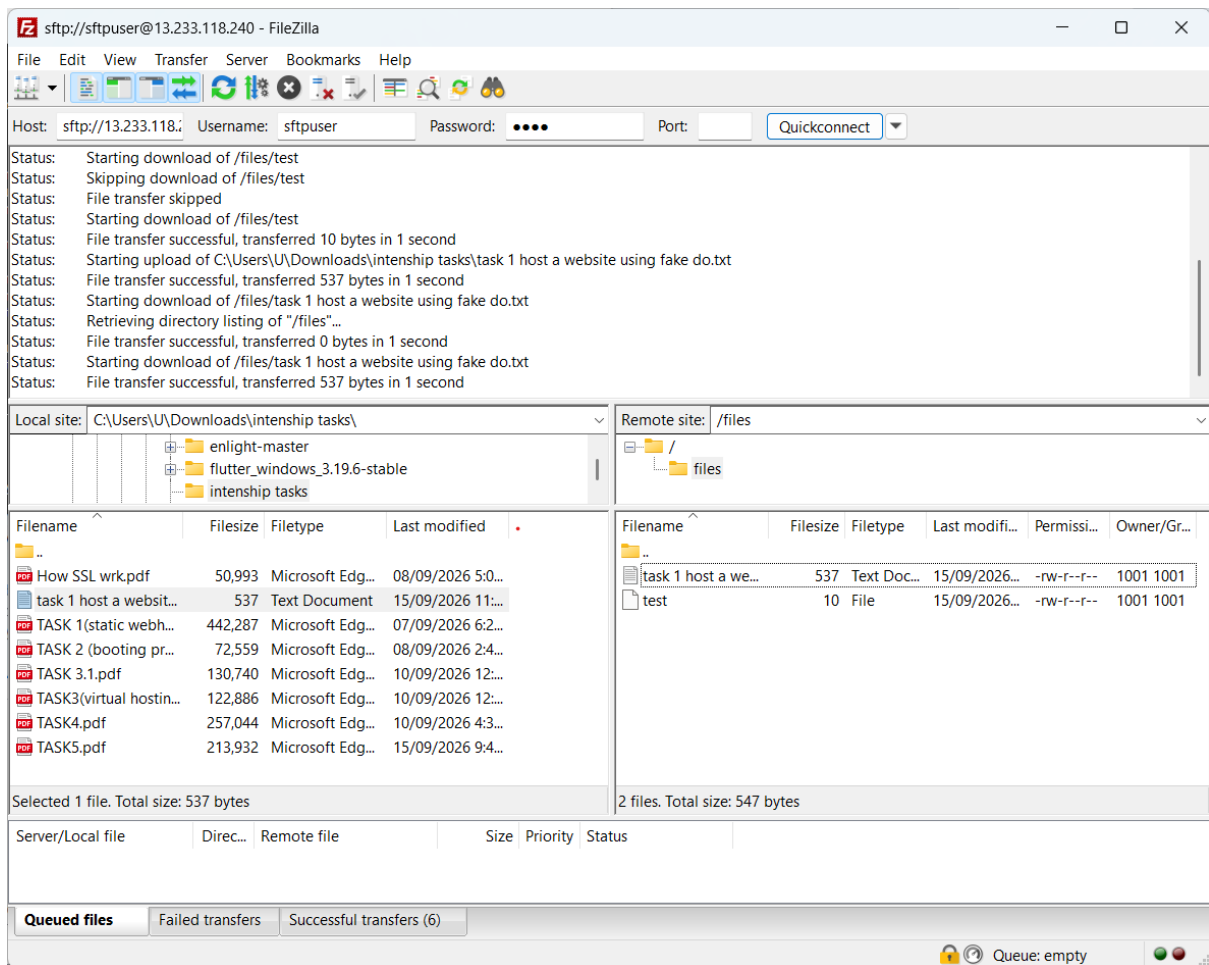
Step 4: Restart SSH

```
#sudo systemctl restart sshd  
  
#sudo systemctl status sshd
```

Step 5: Verify with an SFTP client (FileZilla/Cyberduck)

- **Protocol:** SFTP – SSH File Transfer Protocol
- **Host:** your server IP
- **Port:** 22
- **Username:** sftpuser
- **Password:** as configured

OUTOUT:



Part 2: VSFTPD Setup (Port 21) with Restricted, Secured Users

Step 1: Install vsftpd

```
#sudo apt update
```

```
#sudo apt install vsftpd -y
```

Step 2: Backup default config

```
#sudo cp /etc/vsftpd/vsftpd.conf /etc/vsftpd.conf.bak
```

Step 3: Create the FTP user

```
#sudo useradd /home/ftpuser -s /sbin/nologin ftpuser
```

```
#sudo passwd ftpuser
```

Step 4: Set up the restricted directory

```
#sudo mkdir -p /home/ftpuser/ftp/uploads
```

```
#sudo chown nobody:nogroup /home/ftpuser/ftp
```

```
#sudo chmod a-w /home/ftpuser/ftp
```

```
#sudo chown ftpuser:ftpuser /home/ftpuser/ftp/uploads
```

The chroot root (ftp/) must **not** be owned by the ftp user itself (vsftpd security requirement) — only uploads/ inside it is writable.

Step 5: Configure vsftpd.conf

```
#sudo nano /etc/vsftpd/vsftpd.conf
```

Key settings:

```
anonymous_enable=NO

local_enable=YES

write_enable=YES

local_umask=022

chroot_local_user=YES

allow_writeable_chroot=NO

user_sub_token=$USER

local_root=/home/$USER/ftp


# Passive mode (important behind NAT/cloud instances)

pasv_enable=YES

pasv_min_port=40000

pasv_max_port=40100


# TLS/SSL for security ("very securely")

ssl_enable=YES

rsa_cert_file=/etc/ssl/certs/vsftpd.pem

rsa_private_key_file=/etc/ssl/private/vsftpd.key

force_local_data_ssl=YES

force_local_logins_ssl=YES

ssl_tlsv1=YES
```

```
ssl_sslv2=NO
```

```
ssl_sslv3=NO
```

- chroot_local_user=YES — jails users to their home directory
- pasv_min_port / pasv_max_port — required range for passive FTP data connections through firewalls/cloud security groups
- ssl_enable=YES + force_local_data_ssl/logins_ssl — enforces **FTPS** (encrypted), since plain FTP sends credentials in cleartext

Step 6: Generate a self-signed TLS certificate (or use a real one)

```
#sudo openssl req -x509 -nodes -days 365 \  
-newkey rsa:2048 \  
-keyout /etc/ssl/private/vsftpd.key \  
-out /etc/ssl/certs/vsftpd.pem
```

Step 7: Restrict users to their home directory only (optional whitelist)

```
#echo "ftpuser" | sudo tee -a /etc/vsftpd.userlist
```

In vsftpd.conf:

ini

```
userlist_enable=YES  
userlist_file=/etc/vsftpd.userlist  
userlist_deny=NO
```

This means only listed users can log in at all.

Step 8: Restart vsftpd

bash

```
#sudo systemctl restart vsftpd  
#sudo systemctl enable vsftpd  
#sudo systemctl status vsftpd
```

If on AWS/GCP/Azure, also open these ports (21 + passive range) in the **security group / NSG**, not just the OS firewall.

Step 10: Verify with FTP client

- **Protocol:** FTP with explicit TLS (FTPES) in FileZilla/Cyberduck

- **Host:** server IP
- **Port:** 21
- **Username/Password:** ftpuser
- Client should show a lock icon confirming TLS negotiation, and should be confined to uploads/

OUTPUT:

